Swetha Ashok

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Education

University of California, Berkeley **May 2028**
B.S. Electrical Engineering and Computer Science, Minor in Energy Engineering **GPA: 3.5/4.0**
Relevant Coursework: Circuits & Devices, Data Structures, PCB Design, Computer Architecture, Microfabrication, Microelectronic Devices, Digital Design & FPGAs (Spring 2027), PV Cells (Spring 2027), Intro to Power (Spring 2027)

Skills

Hardware: PCB design and bring-up, ESP32, embedded systems, soldering, debugging, oscilloscopes, digital multimeters, parameter analyzer
Software: C, Python, Java, RISC-V, Github, MS Suite, LaTeX
CAD: SOLIDWORKS, Onshape, Autodesk Fusion 360, KiCAD, LTSpice

Technical Experience and Projects

Electrical Engineer (Current), Solar Project Manager, Calsol - Solar Car Racing Team **Aug 2024 - PRESENT**
- Led a 20+ member team in training for solar module wiring, bypass diode soldering, and PV cell integration.
- Designed integration of the solar array into battery via MPPT, coordinating with electrical and mechanical teams on cross-functional requirements leading to expedited timelines.
- Spearheaded design and hardware build of a mini solar car project, using repurposed PV cells and material.

Project: PCB Retro Synth, Co-Designer and Developer **Fall 2026**
- Delivered a fully functional 12-note analog PCB synthesizer on a 12-week development timeline by collaborating in a 3-person engineering team across design, hardware bring-up and custom firmware.
- Designing schematics and layout with KiCAD, managing BOM to meet performance and budget constraints.

Lab: Microfabrication Technology Course **Aug. 2026 - Dec. 2026**
- Fabricating integrated circuit devices on silicon wafers in a cleanroom environment, executing full-cycle planar processing steps from blank substrate to finished device.
- Conducting characterization and process parameter verification to evaluate device functionality and yield across process steps.

President, Program Manager, FIRST Robotics Competition Team 2643 **Aug. 2020 - May 2024**
- Directed the design, wiring, and testing of multi-sensor robotic systems using SOLIDWORKS, enabling the 50-member team to build functional prototypes and competition ready robots that met electrical and mechanical specifications.
- Streamlined subsystem assembly by maintaining the master CAD and driving synchronous hardware/software milestones.

Leadership and Extracurricular Activities

Executive, Treasurer, Society of Women Engineers **Aug. 2024 - PRESENT**
- Accelerated reimbursement turnarounds by 75% across 100+ active members by managing budget allocations and financial counseling for 30+ distinct programs.
- Organizing logistics and preparing 25 collegiate attendees for industry recruitment through career-prep workshops for WE National and Regional Conferences.

Peer Adviser, UC Berkeley College of Engineering, Student Services **Aug. 2026 - PRESENT**
- Advise prospective and undergraduate engineering students on college requirements, major options, and administrative policies such as course planning and major changes, serving over 100 students to date.
- Serve as the sole hardware-focused EECS peer adviser, delivering specialized guidance on degree planning, course selection, hardware tracks, campus resources, and career pathways.

Executive, Secretary, Institute of Electrical and Electronics Engineers **Aug. 2026 - PRESENT**
- Manages records, supports officer initiatives, and guides strategic decision-making by the Executive Board.
- Bolster new outreach initiatives focused on increasing engagement with community colleges and women in electrical engineering, expanding member participation across chapter activities.